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Department of CSE-Cyber Security(CY & IY)

Assignment Report – I

Course Coordinator	Swathi Hiremaregowdru	Semester/Section	3 sem / IY2A
Course	Object Oriented Programming	Course Code	UE24CS2303
CO Mapped	CO5	Max Marks	10
Issue Date	/ /25	Submission Date	10/11/25



Submitted By

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Topic: Railway Ticket Booking with PNR Generator

COs	CO5	
Allotted Marks for COs	10	
Marks obtained for COs	9	
Submitted Date	Max Marks	Marks obtained
/11/25	10	9

Academic Year: 2025-26 (ODD Semester)

Signature of the Student

Signature of the Course Coordinator

Subject

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1. Introduction

Transportation systems are among the most important infrastructure components of any country. Among them, the railway network plays a vital role in connecting cities, enabling trade, and supporting the daily commute of millions of passengers. As railway operations expanded globally, the process of manual ticket booking and revenue management became inefficient, error-prone, and time-consuming. Hence, the need for automated & digital ticket booking systems became essential.

A Railway Ticket Booking System provides passengers with the ability to reserve trains, select seats & confirm booking without physical paperwork or human dependency. Every booking requires a unique ID number, commonly known as PNR, which stores all essential travel & passenger information. The PNR system is used by Indian Railways & other transportation agencies to manage passengers' details, train numbers, routes, timings, & seat allocations in a centralized database.

This project titled "Railway Ticket Booking with PNR Generator" aims to develop a miniature model of such a system using 'Java' programming language. It demonstrates the concept of generating unique PNR numbers automatically for each booking. The project focuses on logical design, simplicity, of educational demonstration rather than complex enterprise-scale database system.

Additionally, the system highlights the importance of automation in the travel industry, demonstrating how even a basic prototype can save time, reduce manual effort, & improve record management efficiency. It showcases how logical Java code can replicate the core backed process of a ~~much~~ larger real-world system.

2. Objectives

The primary objectives of this project are:

1. To simulate a simple railway ticket booking system using Java.
2. To design & implement a reliable PNR generation mechanism using Java.
3. To demonstrate basic persistent like storage for demonstration.
4. To provide clear documentation & an easy-to-run Java utility for PNR generation.
5. To apply object-oriented programming principles in a practical mini-project.

3. Literature Survey

Online railway reservation system operate at large scale & provide a wide range of services including search, booking, cancellation, seat allocations, & payment integration. These systems are complex & require fault-tolerant database, strict security & high availability.

5. Proposed system

The proposed system is compact prototype that demonstrates PNR generation & booking flow.

Key features include:

- Generation of unique PNR based on train identifier, current date, & a sequence number.
- simple booking flow that allocates sequential seat number per train for demonstration.
- Export capability to CSV from the frontend & an independent Java utility to generate & validate PNR strings.
- This prototype intentionally avoids complex dependencies so that it can be run easily on any developer machine with Java installed.

5. System Architecture / Design

The system follows a 3-layered architecture suitable for small prototypes:

1. Presentation Layer: HTML / CSS / Java - UI that collect I/P & display results.
2. Application Layer: Java utility that encapsulates core PNR generation logic.
3. Data Layer: For the demo, data is stored in browser local storage; for production, a relational database would be used.

The flow can be represented as:

User → Frontend Form → PNR generation logic →
PNR returned → Display to user.

Design Consideration:

- Uniqueness: PNR uses date + sequence to reduce collision.
- Simplicity: No external service required for basic demo.
- Extensibility: Java logic can be integrated into a backend in the future.

6. Modules Description

1. Train Information Module :

- maintains a list of trains (ID, name, etc).

2. Booking modules :

- Accept passenger details & assigns the next available seat number for a train.

3. PNR Generation Module :

- Java class responsible for generating PNR identifier using train ID. & sequence.

4. PNR Checking Module :

- Allows retrieval of booking details by searching with PNR.

5. Data management Module :

- For demo, data is kept in browser local storage. Can be replaced with a database for persistence.

7. Technologies Used

Frontend : HTML, CSS, Javascript (for UI)

Backend : Java (PNR utility)

IDE : Visual Studio Code

Compiler : Java JDK (java/javac)

Platform : macOS / Windows.

8. Implementation & Output Screens.

Steps:

1. Create project folder on Desktop: Railway
2. Open vs code & create two files: .html & .java.
3. For Java code: write the generatePNR method that combines trainId, date, & a sequence number.
4. Compile the Java program using 'javac PNRutils.java' & run using 'java PNRutils'.

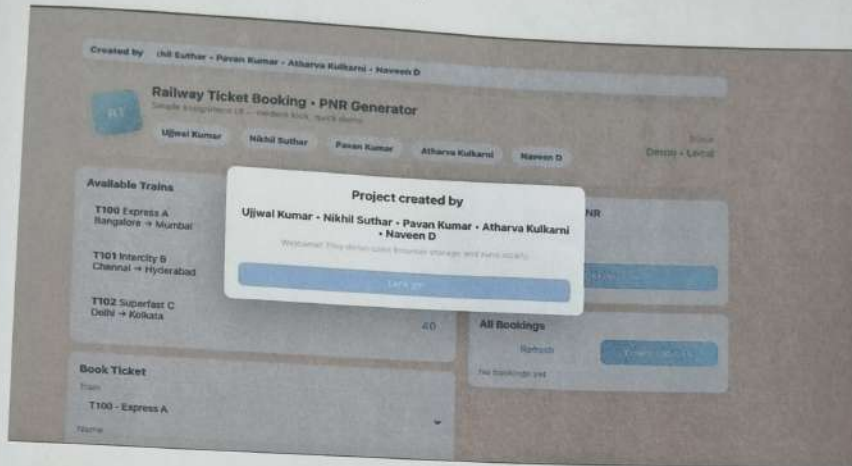
* Sample Output (when running the Java utility):
== Railway Ticket Booking - PNR Generator ==

Generated PNR1: T100-20251102-1001

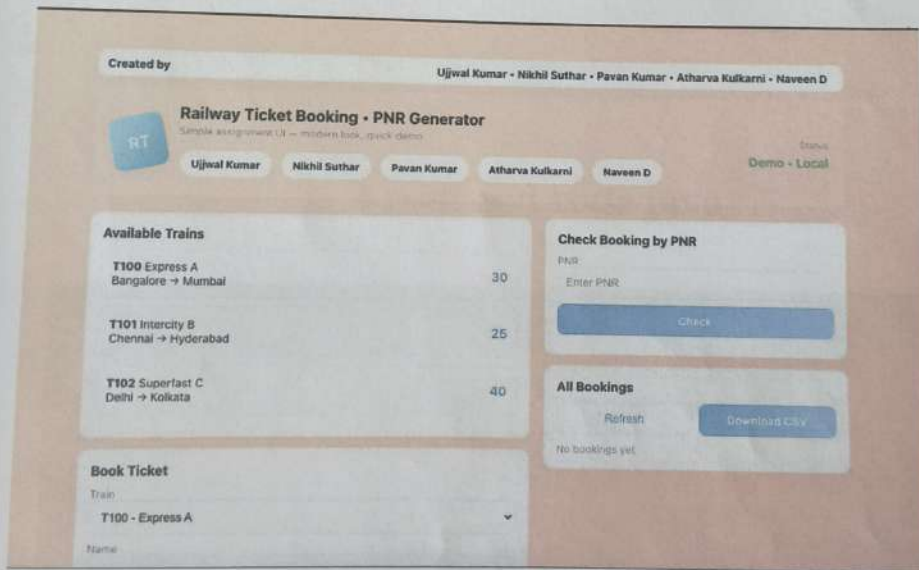
Generated PNR2: T102-20251102-1002

OUTPUT AND CODE SCREENSHOTS

UIUX-1



UIUX-2



Front end code java

[illegible]

Ujjwal Kumar Nikhil Suthar Parsh Kumar Atharva Rukanti Harven D Dimple Local

Available Trains

Train	From	To	Class	Price
T100 Express A	Bangalore	Mumbai	2B	29
T101 Intercity B	Chennai	Hyderabad	2B	29
T102 Superfast C	Delhi	Kolkata	4B	40

Book Ticket

Train: T100 - Express A

Name: Nikhil Suthar

Age: 22 Gender: Male

Book Ticket Clear Bookings

Check Booking by PNR

PNR: T100-20251102-1001

PNR: T100-20251102-1001

Name: Nikhil Suthar - T100 - Seat 1

All Bookings

Refresh View All Bookings

T100-20251102-1001 Nikhil Suthar - 22 - Male T100 Seat 1

SUCCESSFULLY PNR GENERATED-5

```

RailwayProject - ssh - 97x24
Last login: Sun Nov  2 11:43:28 on ttys001
(base) software@Software-MacBook-Air ~ % cd ~/Desktop/RailwayProject
(base) software@Software-MacBook-Air RailwayProject % ls
PNRUtils.class  PNRUtils.java  index.html
(base) software@Software-MacBook-Air RailwayProject % javac PNRUtils.java
(base) software@Software-MacBook-Air RailwayProject % java PNRUtils
Generated PNR: T100-20251102-1001
(base) software@Software-MacBook-Air RailwayProject %

```

JAVA COMPILE TERMINAL

10. Results & Analysis

The prototype reliably generates unique PNR identifiers appended with sequential number. The approach is simple & suitable for academic demonstration.

Limitation: sequence resets when the application restarts unless persisted externally; collisions are unlikely for this demo but a database-backed sequence generator would be recommended for production.

11. Application & Future scope

Applications :

- Educational demonstration for understanding booking workflows & unique identifier generation.
- Prototype for small institutions needing local ticketing solutions.

Future Enhancements :

- Integrate with a relational database for persistence.
- Add authentication & admin features.
- Implement payment gateway integration & notification.